Research Report: Data Envelopment Analysis in Healthcare

Multiple Criteria Decision Support Systems

Màster en Enginyeria de la Seguretat Informàtica i Intel·ligència Artificial (URV)
Màster interuniversitari en Intel·ligència Artificial (UPC, UB & URV)

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Attribute	**Details**
Method Name	DEA in healthcare
Authors and Date	J. Santos, E. Rosário Negas, L. Cavique. (2013) [1]
Versions or extensions of the basic method	Super Efficency DEA [18], Fuzzy-DEA [19], Network DEA [20] and Stochastic DEA [21]
Number of papers on applications found	92 on Scopus and 108 in Web of Science
Is there any survey paper (reference)	Yes, [15] and [16] as the most relevant ones.
Software available for this method	DEAP.2 software [11], EMS 1.3 [12], GAMS Software [13], DEA Solver [14], MaxDEA 7.0 [20]
Usual size of the dataset (As number of DMUs, inputs and outputs)	Small studies: 10–30 DMUs Medium studies: 30–250 DMUs Large studies: 250+ DMUs Number of **input and outputs** can vary generally between 1-12. [17]

30th March 2025

1 Data Envelopment Analysis (DEA)

1.1 Introduction

Data Envelopment Analysis is a mathematical programming technique introduced in 1978 by Charnes, Cooper, and Rhodes [2]. However, its foundations, can be found previously in Farrell's work in 1957 [3], where he first explored the concept of measuring productive efficiency.

Since its first application in agriculture, DEA has been applied in different fields, such as in healthcare, where this report will focus its research, as in one of its first applications by Wilson et al.(2012)[4], who used DEA to monitor the financial performance and technical efficiency of critical access hospitals, combining DEA with logistic regression modeling to assess hospital efficiency.

DEA is considered **non-parametric linear programming method** used to evaluate the relative efficiency of decision-making units (DMUs) by comparing their inputs and outputs to identify the most efficient entities and measure inefficiencies in others [3].

In this context, non-parametric means that **DEA does not assume a predefined mathematical relationship between inputs and outputs**. Unlike parametric methods like regression, which assume a specific mathematical relationship between inputs and outputs, DEA measures efficiency directly from observed data without relying on a predefined formula.

Regarding DMUs, these are entities being evaluated in DEA. Each DMU uses inputs to produce outputs [1]. For instance, in healthcare, a hospital or its departments can be considered DMUs, where inputs like doctors, nurses, and medical equipment are used to deliver patient care and treatments as outputs.

1.2 Explanation of the DEA Method

As previously mentioned, DEA is a method used to measure how efficiently different entities use their resources to produce results, in the field of Healthcare most cases measure efficiency of Hospitals.

In the case of hospitals, each instance is considered a DMU, and each DMU has resources **(inputs)** to produce results **(outputs)**.

For instance, **Hospitals** $\rightarrow$ Inputs: Number of doctors, nurses, beds. Outputs: Number of treated patients, survival rates.

The goal of DEA is to find efficient and inefficient DMUs. It works by creating a frontier to compare with other DMUs.

- **Efficient units** are on the frontier. That is, best DMUs at using inputs to get outputs.
- **Inefficient units** are below the frontier, the DMUs that could do better with the same resources.

To ilustrate how DEA works, we will briefly explain DEA using one of the reference papers that explain the methodology of DEA [1]. In this paper, 12 DMUs that in our field could be Hospitals, are measured with two inputs X_1 & X_2 that could be number of doctrs and number of beds and two outputs Y_1 & Y_2 that could be patients treated and number of drugs used. To simplify the example, we will just use one input X_1 and one output Y_1.

There are two ways for a DMU to reach efficiency: increasing output until it reaches the maximization of output (M.O.), or reducing the input until it reaches the minimization of input (m.i.).

From figure 1 we can observe that the most efficient DMU is DMU 3 as it is the one with the highest slope. If we plot the points from Table 1, we obtain Figure 1 with the 12 observations on a scatter plot.

It is observable that the most efficient one is DMU 3, since the straight line originating at the (0, 0) point towards DMU 3 has the highest slope compared to any of the others. As explained in the paper, this straight line originating at the (0, 0) point towards DMU 3 is called the efficiency frontier, and together with the X-axis, it defines a cone with its vertex at the origin. This cone is referred to as the production possibility set, as it contains all the real data, and according to DEA, only points inside this cone correspond to feasible working conditions based on the best-achieved performance.

DMU	X_1	X_2	Y_1	Y_2
1	4.0	140.0	2.0	28.0
2	5.0	90.0	1.0	22.5
3	6.0	36.0	6.0	12.0
4	10.0	300.0	8.0	60.0
5	11.0	66.0	7.0	16.5
6	8.0	36.0	6.0	12.0
7	9.0	12.0	7.0	6.0
8	5.0	210.0	3.0	30.0
9	5.5	33.0	4.4	5.5
10	8.0	288.0	4.0	72.0
11	10.0	80.0	2.0	20.0
12	8.0	8.0	1.0	4.0

Table 1: Dataset of DMUs with two inputs (X_1, X_2) and two outputs (Y_1, Y_2)

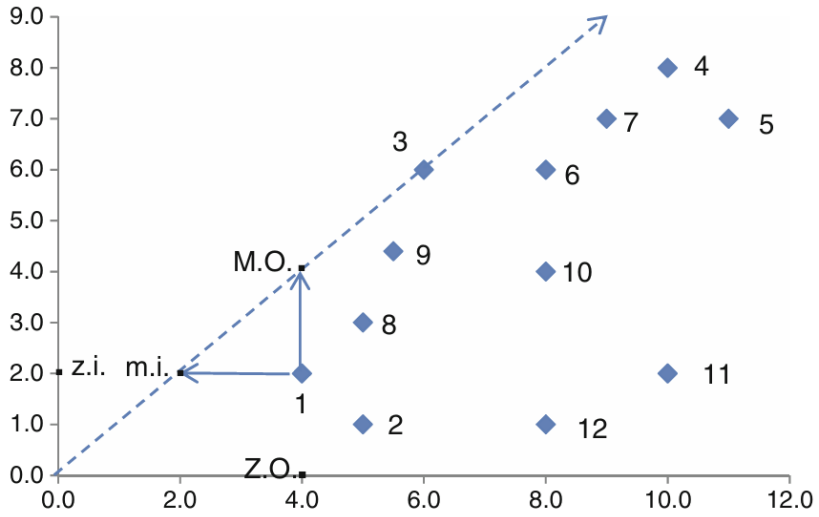


Figure 1: DEA Plot for Table 1: input X_1 and output Y_1

For instance, we will now analyze DMU 1 in more detail. There are two ways for DMU 1 to reach efficiency: either by increasing the output until it reaches the maximization of output (M.O.) or by reducing the input until it reaches the minimization of input (m.i).

The efficiency value is defined as the ratio of the distances:

$$\text{d.m.i.} = \frac{d_1}{d_i} = \frac{0.4}{0.8} = 50\%$$

Similarly, the inefficiency value is defined as the ratio of the distances:

$$\text{d.M.O.} = \frac{d_1}{d_0} = \frac{0.4}{0.2} = 100\%$$

This is the simplest case of DEA with one input and one output. But in the majority of cases DEA extends to more than one input and one output.

Now, we extend the concept to multiple inputs and outputs. When there are multiple inputs $(X_1, X_2, \ldots, X_n)$ and multiple outputs $(Y_1, Y_2, \ldots, Y_m)$, the efficiency of a DMU is determined by finding a set of optimal weights that maximizes the efficiency score.

In multidimensional DEA, the **efficiency score** of a given DMU k is the ratio of the weighted outputs to the weighted inputs. The goal is to find the optimal weights λ_i for inputs and λ_j for outputs such that the efficiency is maximized for each DMU:

$$\text{Efficiency of DMU}_k = \frac{\sum_{j=1}^{m} \lambda_j Y_{jk}}{\sum_{i=1}^{n} \lambda_i X_{ik}}$$

Where:

- Y_{jk} is the output of DMU k for output j,
- X_{ik} is the input of DMU k for input i,
- λ_j are the weights assigned to the outputs,
- λ_i are the weights assigned to the inputs.

To ensure a fair comparison between DMUs, the DEA model also includes constraints. **The weighted outputs for each DMU must not exceed the weighted inputs for any other DMU**:

$$\sum_{j=1}^{m} \lambda_j Y_{jk} \leq \sum_{i=1}^{n} \lambda_i X_{ik} \quad \text{for all DMUs } i$$

Additionally, the sum of the weights for the inputs is typically normalized:

$$\sum_{i=1}^{n} \lambda_i = 1$$

Once all the efficiency scores are calculated for each DMU, the efficiency frontier is formed. The DMUs that achieve the highest efficiency scores (those with a score of 1) are considered the most efficient. These DMUs lie on the efficiency frontier.

In other words, the efficiency frontier is made by identifying the best performing DMUs those that cannot improve their performance without using more resources or achieving more output.

DMUs that lie below this frontier are inefficient. These DMUs are performing worse than the ones on the frontier and may need to improve either by increasing their outputs or reducing their inputs to become more efficient.

So, as a summary, in DEA, **linear programming** is used to calculate the efficiency score for each DMU by finding the optimal weights for inputs and outputs that maximize the ratio of weighted outputs to weighted inputs. The DMU's efficiency score is then obtained from this maximization, with a score of 1 indicating efficiency. After calculating the efficiency scores for all DMUs, those with a score of 1 form the efficiency frontier the most efficient DMUs. DMUs with scores below 1 are considered inefficient and need improvements in performance to reach the frontier.

2 Research Methodology

In this study, we aim to analyse different applications within the DEA method and the related literature to identify common applications and different groups working on this methodology and in which regions. For this analysis, we will restrict the research over the last 10 years, therefore, we will look only on publications after 2015.

We conducted an initial search using three search engines, **Web of Science** and **Scopus**. Despite Data Envelopment Analysis is being commonly referred to as **DEA**, we performed our search using the full name of the method to ensure proper results.

The search parameters used in **Scopus** were:

```
TITLE-ABS-KEY ( data AND envelopment AND analysis AND healthcare ) AND PUBYEAR
> 2014 AND PUBYEAR < 2026
```

For **Web of Science**, we used the following query:

```
Data Envelopment Analysis Healthcare (Topic) and 2015-2025 (Year Published)
```

The search retrieved exactly **500** results from **Web of Science** and **542** from **Scopus**.

Both databases will be used to analyze different trends in the field. While we initially aimed to export and merge the results from both sources, we encountered challenges in matching titles and author names due to minor formatting inconsistencies across databases.

When looking at the publications of applications in DEA in healthcare, a similar approach was used. To find the publications on applications, we modified our search to add applications.

The search parameters used in **Scopus** were:

```
(TITLE-ABS-KEY ( data AND envelopment AND analysis AND healthcare ) AND TITLE-ABS-KEY
( application ) ) AND PUBYEAR > 2014 AND PUBYEAR < 2026
```

For **Web of Science**, we used the following query:

```
Data Envelopment Analysis Healthcare (Topic) and 2015-2025 (Year Published) and
Application (Topic)
```

The search retrieved exactly **108** results from **Web of Science** and **92** from **Scopus**. In this case, Scopus resulted in less publications than WoS.

3 Analyzing DEA Research in Healthcare

Although the increasing demand for healthcare resources is not solely driven by an aging population and rising life expectancy[5], assessing the efficiency of healthcare services remains essential. DEA applications have expanded across various domains, including hospital benchmarking, physician performance evaluation, and cost-effectiveness analysis. This study aims to explore research trends in DEA applications within healthcare by analyzing data from Web of Science and Scopus, identifying key authors, influential institutions, leading countries, and thematic trends.

After an initial search, the most relevant papers found in Web of Science are presented in Table 2. From this table, several key insights emerge, such as the prominence of research on China's efficiency in resource allocation and healthcare services, with two of the five studies focusing on this topic. Additionally, two papers explore the efficiency of healthcare institutions, while another examines decision-making under uncertainty.

3.1 Trends in DEA Research

In this subsection, the trends in DEA research will be explored, focusing on the number of publications and citations over time to understand how interest in this area has developed. The most influential authors will be highlighted, along with the types of publications, such as journal articles, conference papers, and books. Also, the research areas where DEA has had the most impact will be identified.

As shown in Figure 2, from 2015 to 2024, the number of publications has followed a positive trend, starting with no more than 30 publications per year in 2015 and 2016, and increasing to over 80 in 2024. Similarly, the number of citations has increased significantly, from fewer than 100 in 2015 to more than 1,300 in 2024. However, since 2025 is not yet complete, the data for this year may be misleading, as it shows what appears to be a negative trend after 2024. No conclusions can be drawn regarding this potential decline, as the year is still ongoing.

Article	2021	2022	2023	2024	2025	Average	Total
The use of Data Envelopment Analysis (DEA) in healthcare with a focus on hospitals [6]	49	62	47	51	6	37.29	261
China's regional sustainability and diversified resource allocation: DEA environmental assessment on economic development and air pollution [7]	21	19	17	18	3	16.36	180
Multi-expert performance evaluation of healthcare institutions using an integrated intuitionistic fuzzy AHP&DEA methodology [8]	28	17	11	6	1	14.44	130
A cross-efficiency fuzzy Data Envelopment Analysis technique for performance evaluation of Decision Making Units under uncertainty [8]	21	6	10	6	1	11.36	125
Evaluation on equality and efficiency of health resources allocation and health services utilization in China [9]	14	30	19	20	2	13.44	121
All Articles	823	1138	1100	1347	222	543.27	5976

Table 2: Most Cited papers regarding DEA in healthcare from 2021 to 2025

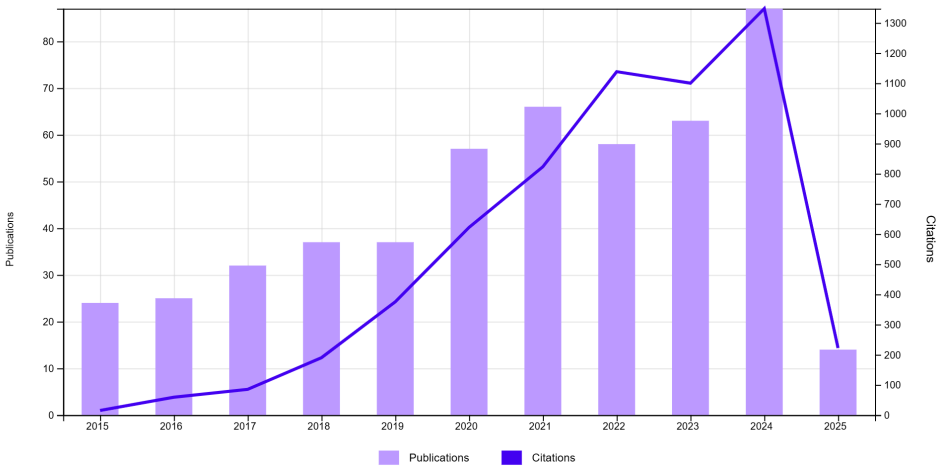


Figure 2: Evolution of publications and citations of DEA in healthcare research from 2015 to 2025

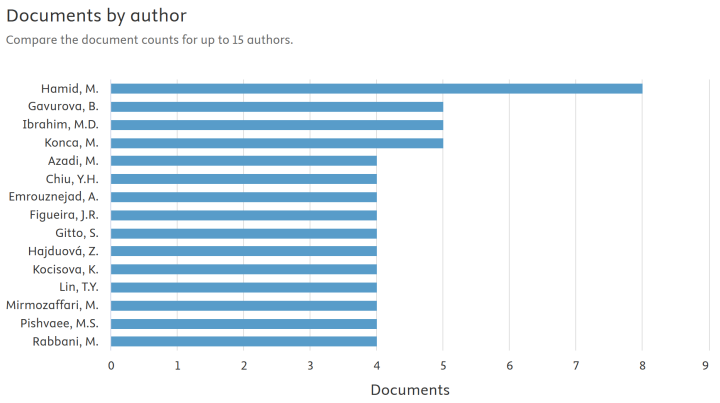


Figure 3: Authors with more publications in DEA in Healthcare since 2015

Regarding the authors, Figure 3 highlights a clear leader in the field: Hamid, M. from the University of Tehran, who has contributed 8 publications since 2015. Following Hamid, there are several other authors, including Gavurova, B. (Technická Univerzita v Košiciach, Košice, Slovakia), Ibrahim, M.D. (HCT-Sharjah Campus, Sharjah, United Arab Emirates), and Konca, M. (Çankiri Karatekin Üniversitesi, Çankiri, Türkiye), each with 5 publications.

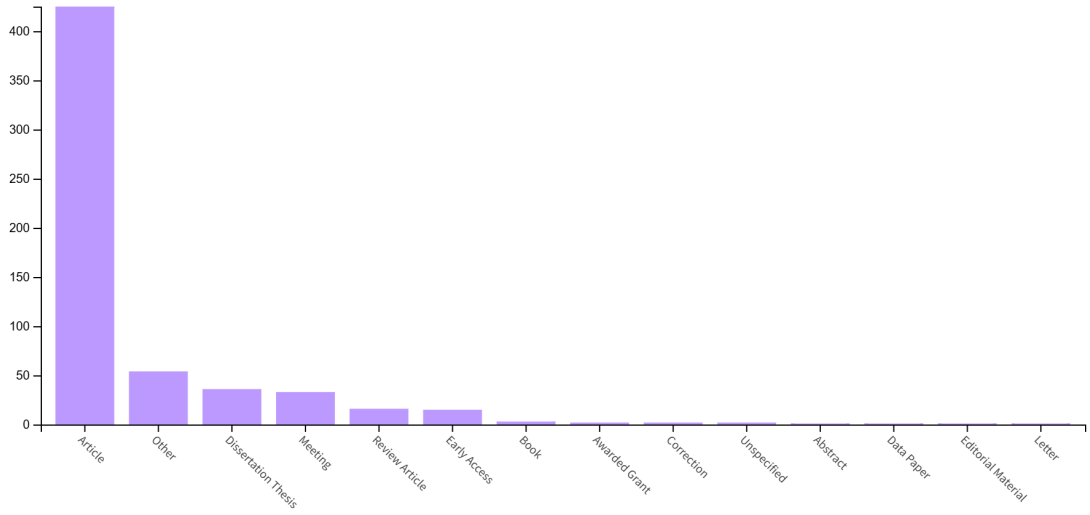


Figure 4: Type of publications in DEA Healthcare since 2015

Figure 4 shows the distribution of publications in the period of 2015 to 2025, where the majority of documents in DEA research are journal articles, followed by less common types such as dissertation theses, which are in-depth research projects typically completed at the PhD level. Other notable document types include meetings, review articles, early access publications, and books. However, less frequent document types, like Corrections and letters, appear rarely in the research.

Field: Open Access	Record Count	% of 500
All Open Access	270	54.000%
Gold	200	40.000%
Gold-Hybrid	28	5.600%
Free to Read	16	3.200%
Green Published	173	34.600%
Green Accepted	19	3.800%
Green Submitted	28	5.600%
Dissertation	1	0.200%

Figure 5: Access

Regarding the type of access of the DEA research documents, the majority are available through open access, with 54% of the total records being classified as open access. Among these, 40% are published under the "Gold" open access model, where the authors or institutions pay fees to make the articles freely accessible to the public.

A smaller proportion, 5.6%, falls under the "Gold-Hybrid" model, where only certain articles in subscription-based journals are made open access. Additionally, 3.2% of the publications are marked as "Free to Read," meaning they can be accessed without any subscription or paywall, although this is typically for a limited time. The "Green" model, shows that 34.6% of the articles are Green Published, 3.8% are Green Accepted, and 5.6% are Green Submitted. Only a very small percentage, 0.2%, consists of dissertation theses.

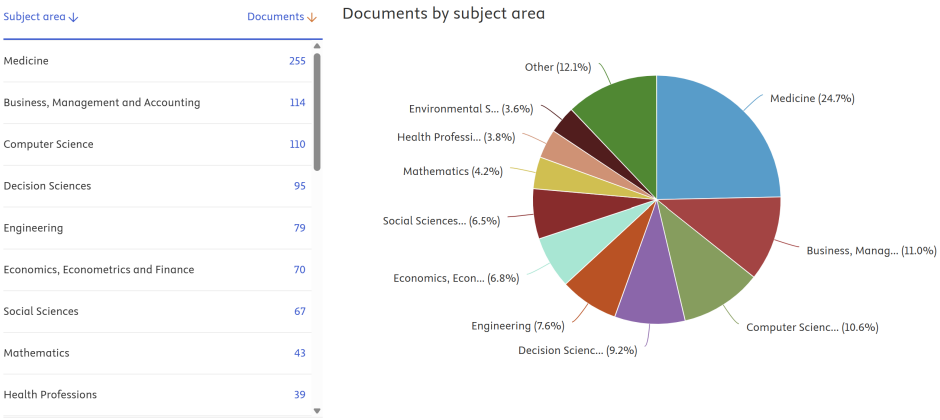


Figure 6: Subject Area

Regarding the Subject area of the publications, DEA in healthcare research is most prominently applied in medicine and health professions, accounting for a combined 28.5% of publications. This dominance is expected, as the search papers are under DEA in healthcare, which mostly include publications used to evaluate hospital efficiency among others. Within this search of DEA in healthcare, publications are also classified as business management (11%), and in computer science (10.6%). Additionally, 9.2% of publications focus on decision sciences, aligning with DEA's fundamental role in efficiency measurement. Other fields, such as engineering, economics, social sciences, and environmental sciences, each contribute less than 8% of DEA research in healthcare, reflecting its broad yet healthcare-centered application.

3.2 Geographical Analysis

The countries with the highest number of publications on DEA in healthcare are led by China, which has contributed over more than 80 publications since 2015. Following China, Iran ranks second with more than 70 publications, and the USA comes third with 64 publications. After these top three, there

is a significant gap, with India, Turkey, and the UK each publishing fewer than 40 articles. Several other countries, including Taiwan, Italy, Spain, and Canada, have published fewer than 30 papers each.

When it comes to continents, Asia clearly leads the way in DEA research within healthcare, with China and Iran contributing the most publications. China's strong presence shows its growing focus on healthcare and resource allocation studies, while Iran's significant number of publications reflects an increasing interest in efficiency analysis in healthcare in the Middle East. The USA is also a key player, but the smaller number of publications from countries like the UK and Canada suggests that Europe and North America might be focusing on other areas of research.

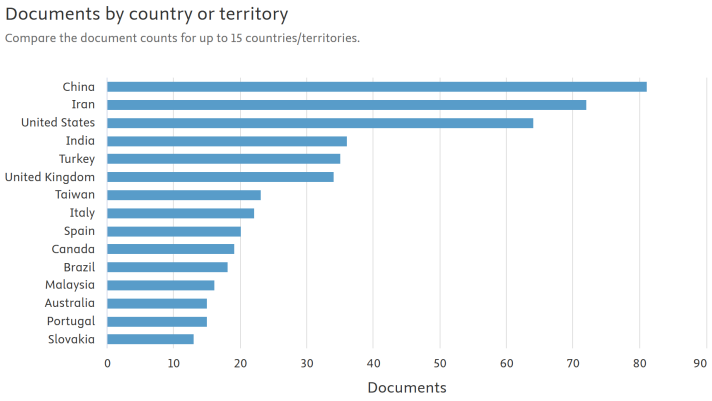
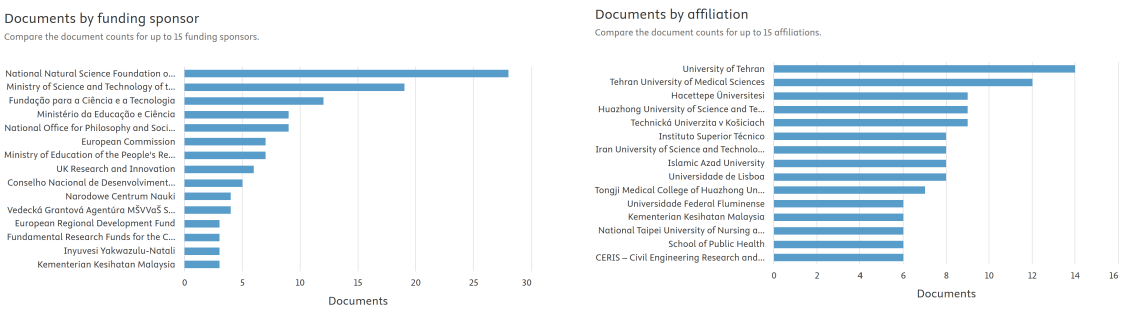


Figure 7: Countries with most publications in DEA in Healthcare since 2015

When analyzing the affiliated universities of the publications, as seen in Figure 8b, the University of Tehran stands out as the most influential institution, with 14 publications related to DEA in healthcare, showing Iran's leadership in the field. Closely wollowed by another university from Teheran, the Tehran University of Medical Sciences, contributing 12 publications, reinforcing the strong presence of Iranian universities in this area. Then, Hacettepe Üniversitesi from Turkey, Huazhong University of Science and Technology from China, and Technická Univerzita v Košiciach from Slovakia each contribute 9 publications, reflecting the growing importance of these countries in DEA research.

These universities align with the countries that dominate DEA research in healthcare. Iran's universities, are clearly leading contributors, reflecting Iran's strong focus on healthcare efficiency. Similarly, universities from China, Turkey, Slovakia, and Portugal also play significant roles, demonstrating the global nature of this research.

Regarding funding sponsors, as seen in Figure 8b the National Natural Science Foundation of China leads the way with 28 contributions, followed by the Ministry of Science and Technology of the People's Republic of China with 19. Other sponsors include Fundação para a Ciência e a Tecnologia with 12, Ministério da Educação e Ciência with 9 publications, the latter two from Portugal. From this analysis we can observe a clear alignment between the funding organizations and the universities working on DEA research in healthcare, particularly in the countries of China, Portugal, and Iran.



(a) Sponsor

(b) Universities

Figure 8: Sponsor and Universities

Select All ☐	Field: Languages	Record Count	% of 500
☐	English	472	94.400%
☐	Portuguese	16	3.200%
☐	Turkish	6	1.200%
☐	Spanish	4	0.800%
☐	Korean	3	0.600%
☐	Greek	1	0.200%

Figure 9: Languages

Finally, when analyzing the language of these publications, we observe that English is by far the most frequently used language. Additionally, Portuguese, Turkish, and Spanish appear, along with Korean and Greek. As shown in Figure 9, a key insight is that despite China having the highest number of publications, these are not primarily in Chinese. Similarly, Iran, the second-leading country in terms of publications, does not predominantly publish in its native languages. Instead, both countries prefer to publish in English.

4 Aplications

4.1 Aplication 1: Has the Efficiency of China's Healthcare System Improved after Healthcare Reform? A Network Data Envelopment Analysis and Tobit Regression Approach

The study selected for this first application is titled "Has the Efficiency of China's Healthcare System Improved after Healthcare Reform? A Network Data Envelopment Analysis and Tobit Regression Approach", conducted by Gong et al. (2019) and published in the International Journal of Environmental Research and Public Health [20]. This paper presents a thorough efficiency analysis of China's healthcare system in the years following the major national healthcare reform introduced in 2009.

In 2009, China launched a reform of its healthcare system with the goal of providing universal basic healthcare to its population. The reform focused on expanding insurance coverage, improving access to services, increasing funding, and restructuring public hospitals. Despite substantial increases in healthcare spending and insurance coverage, questions remained regarding the actual efficiency of healthcare service delivery and outcomes.

The study by Gong et al. addresses this by applying a Network DEA model, which differs from traditional DEA in that it takes into account the internal structure of the healthcare system. Rather than treating the entire system as a "black box," the network model allows for decomposition into two sequential stages:

- **Stage 1:** Resource allocation and healthcare service delivery.
- **Stage 2:** Health outcome production based on services delivered.

This approach provides more insight into where inefficiencies occur and what components of the healthcare system contribute most to them.

The authors analyze 30 provinces in China (excluding Tibet due to lack of data) from 2009 to 2016, using annual panel data obtained from national health statistics yearbooks and government databases. The DEA model is input-oriented and assumes constant returns to scale (CRS). The study also performs a Tobit panel regression to examine external factors influencing the overall efficiency scores.

The Network DEA model uses the following structure:

- Inputs (Stage 1):

Number of health technicians per 1,000 persons.

Number of hospital beds per 1,000 persons.

Per capita total health expenditure.

- Intermediates (Outputs of Stage 1 and Inputs of Stage 2):

  Number of outpatient visits.

  Number of inpatient visits.

  Average inpatient days.

- Outputs (Stage 2):

  Maternal mortality rate.

  Perinatal mortality rate.

  Life expectancy.

All variables are standardized to ensure comparability across provinces and years.
Also, the tools used are MaxDEA 7.0 software for DEA computation and Stata SE 15 for Tobit regression.

The findings of the DEA research provide important information on the performance of China's healthcare system following the 2009 reform. One of the most significant discoveries is that no province reached 100% efficiency over the years investigated. While some provinces were fully efficient in specific years, the total number of totally effective provinces decreased after 2009. Nonetheless, average overall efficiency scores have gradually improved over time, notably after 2012. This trend demonstrates that, despite inconsistent progress, the healthcare reform began to have a positive impact a few years after its implementation.

When the two stages evaluated in the network DEA model are compared, an apparent trend appears. Between 2009 and 2011, the second stage, which focused on health outcomes like maternal mortality, perinatal mortality, and life expectancy, had higher average efficiency scores than the first stage, which addressed resource allocation and service delivery. However, after 2012, the first stage improved significantly, most likely as a result of the universal health insurance coverage achieved in 2011. This increase in access appears to have enabled more people to use healthcare services, hence improving service efficiency. By 2016, however, the second stage's efficiency had once again surpassed that of the first, presumably reflecting increases in overall population health indicators such as life expectancy.

There were also significant differences in efficiency among provinces. The more developed eastern provinces, including as Beijing, Tianjin, and Guangdong, have higher efficiency rankings. This is due to improved infrastructure, more qualified healthcare workers, and more effective management of healthcare resources. In contrast, western provinces tended to be less efficient, highlighting regional differences in healthcare availability and investment. Interestingly, some provinces, such as Guangxi and Guizhou, were able to achieve complete efficiency in specific years, implying that targeted regional approaches or unique local conditions may have contributed to success in those cases.

In terms of temporal evolution, some provinces, such as Beijing, showed consistent and steady improvement year after year, indicating continuous investment and policy success. Other locations, however, demonstrated irregular or static developments, emphasizing the different impact of reforms and the potential involvement of local governance and implementation strategies.
To determine the causes resulting in efficiency fluctuations, the study used a random-effects Tobit regression analysis with DEA efficiency ratings as the dependent variable. This regression revealed many statistically significant results. Among the positive factors, being in a more developed location (e.g., eastern or central China) was related with higher efficiency.

On the other hand, multiple variables have a negative correlation with healthcare efficiency. Surprisingly, provinces with higher rates of higher education enrollment per 100,000 people had lower efficiency rankings. The authors speculate that this could be related to increased demand and strain on large hospitals in more educated provinces, which are frequently more centralized and overcrowded. Social health expenditure, which includes non-governmental sources such as insurance funds, was also found

to be negatively associated with efficiency, probably due to moral hazard and overutilization of services. Furthermore, the number of public hospitals in a province was shown to be inversely associated to efficiency, which the study attributes to potential service duplication, fragmentation, and fee-for-service incentives that may result in unnecessary treatments and waste of resources.
Interestingly, variables such as GDP per capita, personal health expenditure, and the number of private hospitals were shown to have no statistically significant effect on efficiency. This shows that more economic development or out-of-pocket spending does not always result in more effective healthcare delivery.

In conclusion, the study finds that, while healthcare efficiency in China has increased since the reform, the improvements have not been consistent across the country. The improvements are primarily due to better service delivery rather than improved health outcomes. Based on these data, the authors propose various policy changes. First, they underline the importance of strengthening the primary care system, which can help to prevent unnecessary hospital visits. Second, a hierarchical diagnosis system is advocated to guide patients to appropriate levels of care and reduce overcrowding in big hospitals. Third, the healthcare payment system must be significantly reformed, moving away from fee-for-service models and toward value-based methods that prioritize quality and efficiency over volume.
Overall, this study provides a strong example of how advanced DEA modeling, especially in its network form, can be effectively applied to evaluate the impact of large-scale healthcare reforms. By breaking down the production process into stages and integrating socioeconomic data through regression analysis, the study delivers detailed, actionable insights that are relevant for policymakers and healthcare administrators aiming to optimize system performance.

4.2 Aplication 2: Evaluating Health System Efficiency using Data Envelopment Analysis: A case of Indian Healthcare System

This second application, developed by Coutinho, Prasad, and Prabhudesai (2021) [22], examines the efficiency of India's healthcare system using Data Envelopment Analysis (DEA) to benchmark India against its peers in the BRICS group as well as a broad set of OECD countries. The study is motivated by India's unique position as a low-spending, resource-constrained health system that still achieves noteworthy health outcomes. With rising demand and limited inputs, understanding how efficiently India converts healthcare investment into health outcomes becomes essential.

The main aim of the study is twofold: first, to evaluate India's relative efficiency compared to BRICS countries (Brazil, Russia, India, China, and South Africa); second, to assess India's standing among 36 OECD countries. The authors explore whether India lies on the efficiency frontier, and whether it makes better use of limited inputs than wealthier nations. This comparison provides both diagnostic insights and best-practice references to guide future policy decisions.

The DEA model uses countries as Decision Making Units (DMUs), applying both output-oriented and input-oriented models to highlight different aspects of efficiency. Two main DEA variants are used:

- Super-efficiency Output-Oriented DEA with Variable Returns to Scale (VRS): to rank countries in terms of how well they maximize outcomes.
- BCC Input-Oriented DEA: to assess how efficiently countries minimize inputs for a given level of health outcomes.

The analysis uses 2018 data, sourced primarily from the OECD and WHO databases. The inputs and outputs were carefully selected based on their relevance to macro-level healthcare efficiency, especially for a developing country context.

- Inputs:

  Total health expenditure per capita (PPP USD).

  Number of doctors per 1,000 inhabitants.

  Number of nurses per 1,000 inhabitants

  Number of hospital beds per 1,000 inhabitants

- Outputs:
  - Life expectancy at birth
  - Infant survival rate
  - Maternal survival rate

These output measures—particularly infant and maternal survival—are well-suited for developing nations, capturing critical dimensions of population health that reflect the overall effectiveness of the health system.

India's performance among the BRICS nations is paradoxical. In the super-efficiency model, India scores last (5th), indicating that it is less efficient than its counterparts at maximizing health outcomes. However, in the input-oriented model, India ranks top, indicating that it generates rather strong results despite the extremely limited resources it uses. Despite having the lowest health spending and the lowest density of physicians, nurses, and hospital beds, India performs favorably in terms of outputs. This dual perspective emphasizes that India is not necessarily underperforming; rather, it is under-resourced.

When compared to OECD countries, India's relative efficiency is much more apparent. In both DEA models, India is on the efficiency frontier, alongside Canada, Japan, Korea, Spain, and Israel. Many high-spending countries, including the United States, Germany, France, and numerous Central European countries, fall below the threshold, showing inefficiencies despite significant investment. This indicates that increased spending does not always lead to better results.

The study looks deeper into systems from nations with high rankings, such as Canada, Japan, and South Korea. These countries provide various kinds of universal health care and regulatory frameworks that have been successful in increasing results while minimizing costs. Canada, for example, offers national insurance through decentralized provincial administration, assuring mobility and fair access. Korea made significant advantages by integrating its insurance schemes and enforcing tight drug reimbursement standards. Japan stands out not only for its huge spending, but also for its strict regulation of generics and emphasis on branded quality, which have been combined with universal coverage since 1960.
In contrast, the U.S. ranks low despite spending the most per capita, illustrating that fragmented systems, lack of universal coverage, and market-driven models can lower efficiency.

While India makes good use of its current resources, it falls behind in terms of health outcomes, owing mostly to underinvestment. The system is extremely restricted, with just 0.78 doctors and 1.5 nurses per 1,000 people and a per capita health budget of only USD 208.77. The authors claim that boosting these inputs, together with strategic reforms, will greatly enhance India's performance not only in efficiency but also in absolute health indicators.

This study shows how DEA can be used to influence resource allocation and change in impoverished countries. It defines India's existing efficiency as high but unstable, with persistent underinvestment preventing meaningful increases in life expectancy or mortality reduction. Policies such as expanding universal insurance, regulating medicine pricing and prescriptions, and increasing public infrastructure investment are proposed, drawing inspiration from high-performing Asian systems.

5 Conclusions & Personal evaluation

This study provided a thorough and broad exploration of Data Envelopment Analysis (DEA) and its function for evaluating healthcare efficiency. Starting with its basic foundations, DEA evolved into a diverse and powerful tool for comparing the performance of entities that use different inputs to generate multiple outputs. One of the most useful features of DEA is that it does not require predefined functional relationships, making it useful for complex systems such as hospitals and national health services where interactions between resources and outcomes are difficult to represent accurately.

In analyzing the method, we gained an in-depth understanding of how DEA calculates efficiency in relation to a best-practice frontier, and how this frontier distinguishes between high-performing units

and those that could be improved. Understanding how DEA can be tailored to different evaluation requirements needs an understanding of several model types, such as input-oriented, output-oriented, and variable or constant returns to scale. Just as important was understanding how the selection of inputs and outputs influences the analysis, emphasizing the need of contextually appropriate data in efficiency studies.

The literature review component was very eye-opening. It highlighted how widely DEA has been adopted across countries and institutions, particularly in China, Iran, and India, which are making significant contributions to healthcare efficiency research. The global expansion of applications demonstrated DEA's adaptability to many healthcare contexts, whether well-resourced or under-resourced, and how it helps both strategic decision-making and academic research.

Moreover, the analysis of real-world applications reaffirmed DEA's practical value. In China, DEA was used to assess the long-term effects of systemic healthcare reform. The use of a network model provided additional depth, illustrating how changes in service delivery and health outcomes may not necessarily occur in parallel. The use of regression analysis to assess external variables provided an important layer of complexity and demonstrated how DEA findings could encourage targeted policy proposals.
The Indian case showed a completely different but equally convincing perspective: how a low-input system can still be efficient. Despite limited financial and human resources, India has crossed the efficiency threshold when compared to both BRICS and OECD nations. This discovery defied intuitive beliefs and highlighted how DEA captures relative performance rather than absolute success, making it an important tool for evaluating systems in developing contexts.

From our point of view, this project has been a valuable opportunity to deepen our understanding of DEA, both as a methodological framework and as a practical tool for analyzing efficiency in complex systems like healthcare. Working through the literature, collecting data, and reviewing specific applications allowed us to engage with DEA beyond its theoretical definitions. We were able to see how the method is used in real contexts, what types of decisions it can support, and how its outcomes can be interpreted and used to inform policy or management.

Throughout the process, we became more aware of how critical the design of a DEA model is, particularly in selecting appropriate inputs and outputs, choosing between input or output orientation, and understanding the implications of scale assumptions. These elements are not just technical decisions; they directly affect the results and their validity. This highlighted for us the importance of careful methodological thinking when applying quantitative tools in real-world scenarios.

Overall, the study improved our abilities to evaluate and use analytical approaches in a thorough but practical manner. It also highlighted the value of DEA as a tool for making data-driven decisions. While DEA may not provide conclusive answers, it does provide a solid framework for evaluating performance gaps and finding potential for improvement, which is especially useful in industries such as healthcare, where effective resource utilization can have a major impact.

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